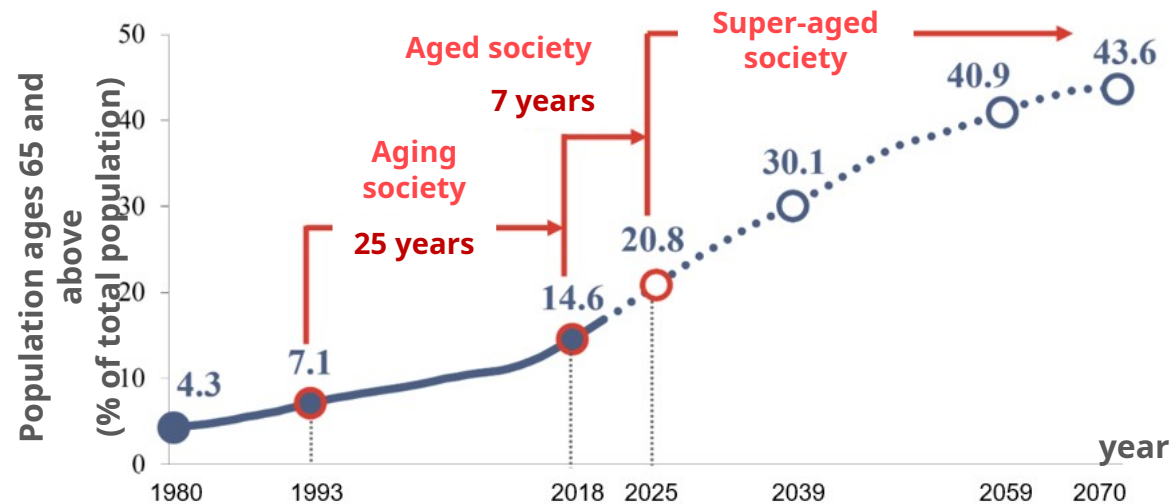




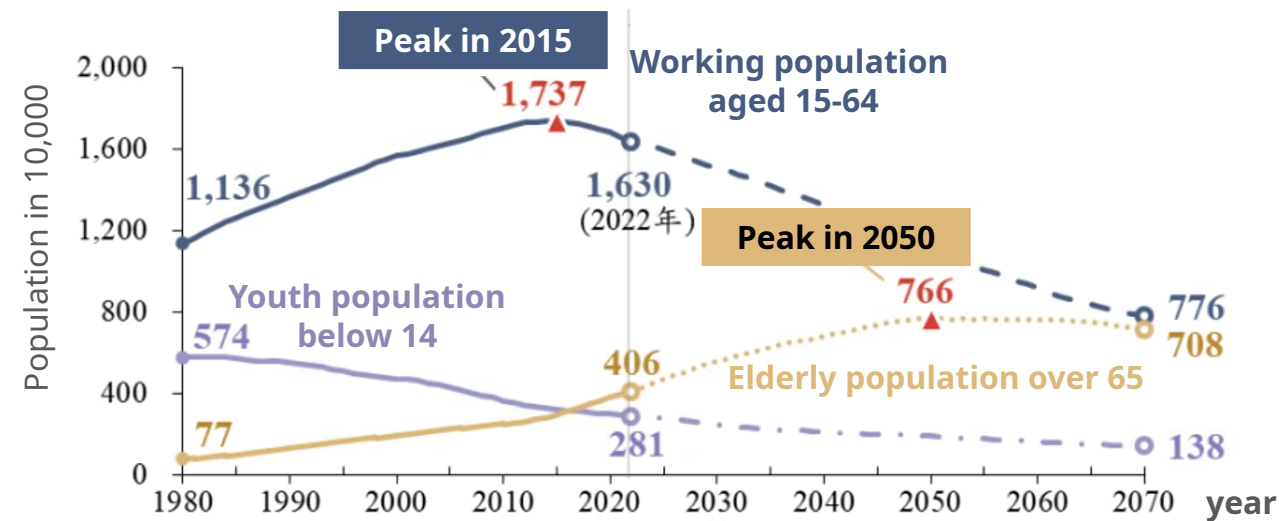
Taiwan Will Become a Super-Aged Society by 2025

- Taiwan became an aging society in 1993, became an aged society in 2018, and is projected to **become a super-aged society by 2025**.
- The proportion of the elderly population will continue to increase and is expected to exceed 30% by 2039, reaching 43.6% by 2027.

- The population aged below 14 began to fall in 1984, and aged 15-64 began to fall in 2016.
- The population aged 65 and over is projected peaked by 2050 and slightly reduce.
- It is projected that by 2065, the population aged 65 and over will increase to 5 times that of the population under 14.



Population projection by the National Development Council (NDC)



2022 Population projection by the National Development Council (NDC)

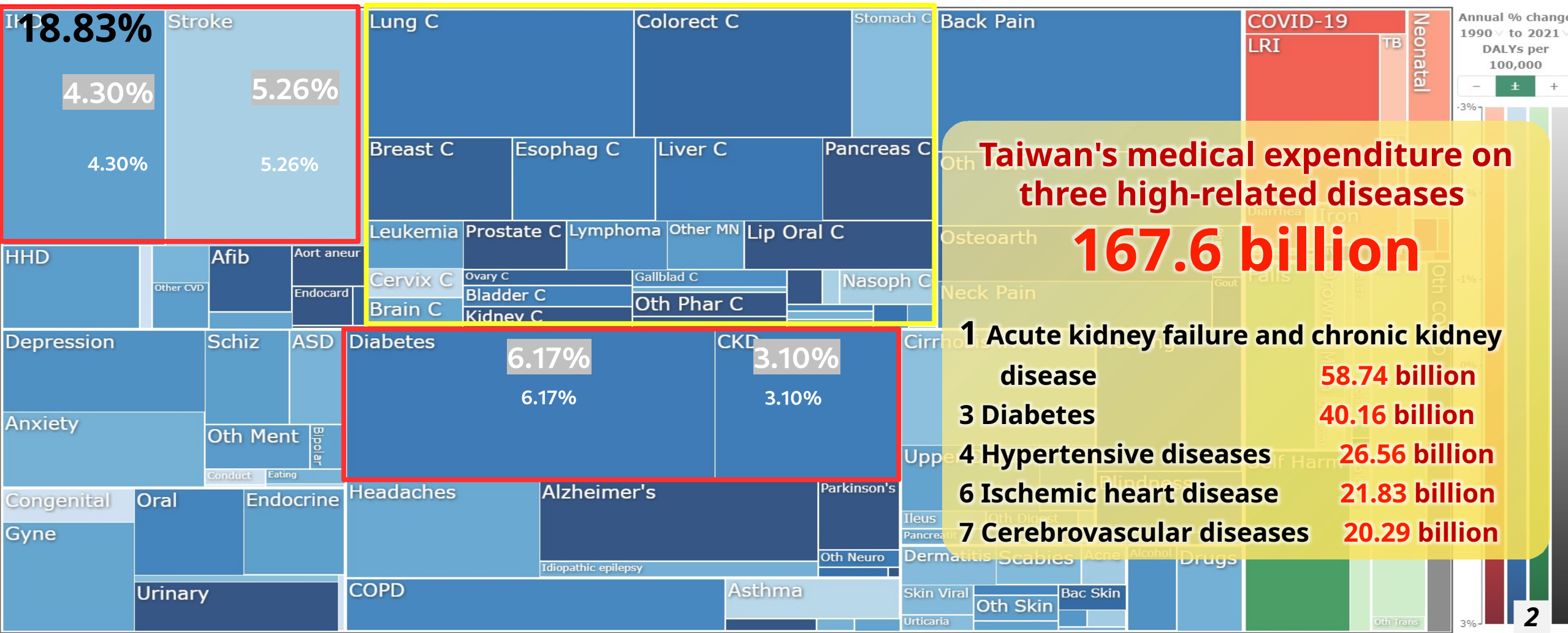


DALYs by Cause in Taiwan, 2021

The disease burden of NCD in Taiwan is about

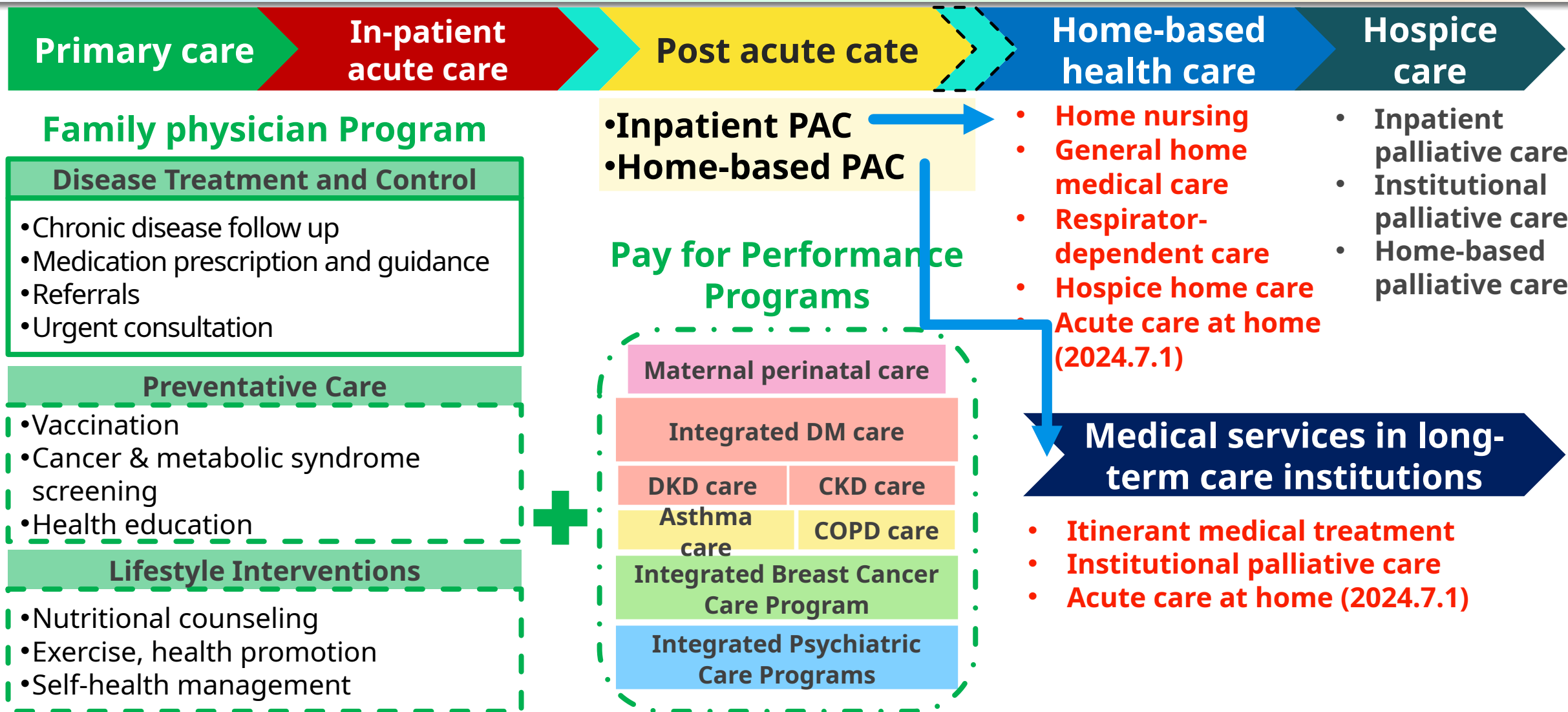
Global Burden of Disease Study 2021

data origin: <https://vizhub.healthdata.org/gbd-compare/>





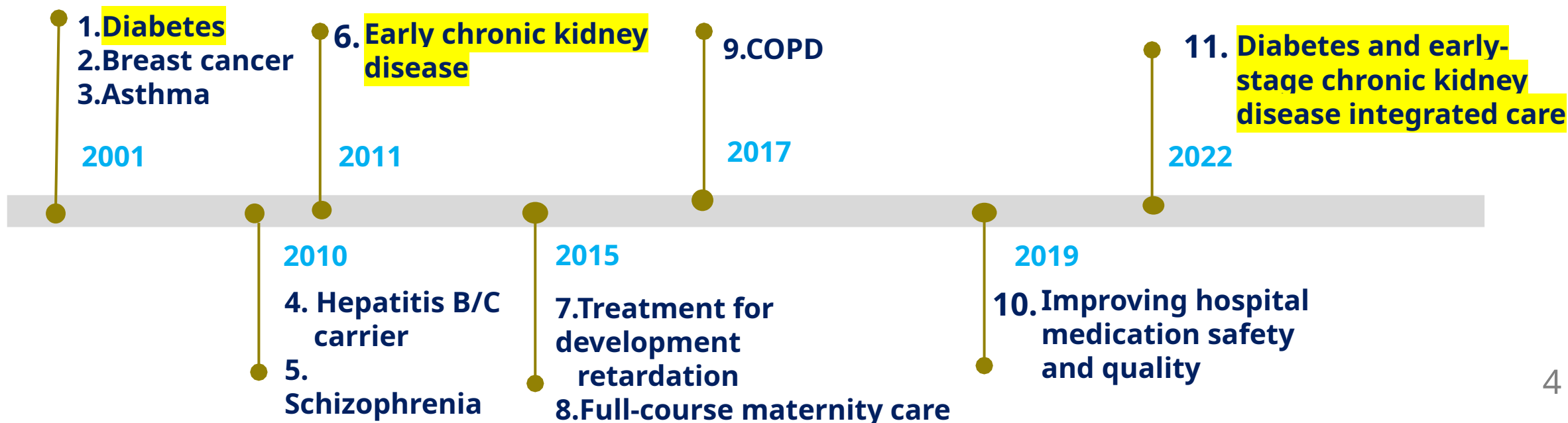
NHI Programs for Whole Life Course Care





The Driving Timeline of the P4P Programs

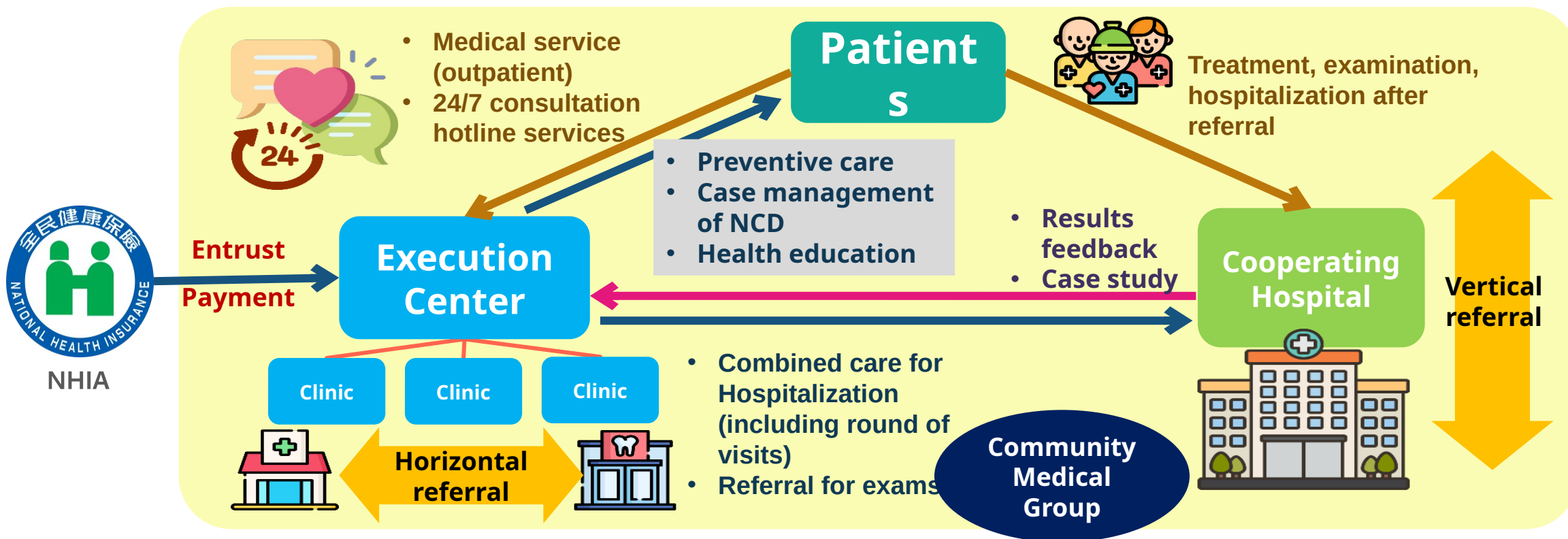
- Since 1990, the NHIA have gradually implemented pay-for-performance plans for disease with large population, high expenditure and needs for quality improvement, such as **diabetes and early chronic kidney**.
- In addition to the fee-for-service, the NHIA provides additional management care fees and quality bonuses which encourage medical institutions to proactively and regularly monitor patients' conditions, slow disease progression, and improve the quality of care.





Family Physician Program Launched in 2003

- Encouraging the formation of “**Community Medical Groups**”—coalitions of five or more local clinics and referral hospitals, to offer the holistic care for NCD patients.
- Priority is given to **elderly individuals, chronic disease patients, and those with high utilization**, who are assigned to clinics participating in the program for health management.



Challenges and Dilemmas

Insufficient coverage and primary physician's participation

- ✓ Only 25% of the eligible population is enrolled.
- ✓ Clinic participation rate is approximately 50%.



The number of patients with comorbidities is increasing annually along with rapid population aging

- ✓ In 2022, approximately 17.4% of the population is aged 65 or older.
- ✓ It will be over 20% by 2025.



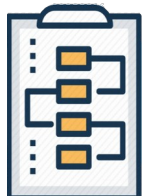
Quality of care for chronic disease patients needs improvement

In 2022, among the plan members, there were 1.2 million patients with diabetes mellitus (DM) or chronic kidney disease (CKD), with only 640,000 individuals receiving care under the P4P program



The goal of comprehensive integrated care has not yet been achieved

The Family Physician plan and P4P program operate independently, providing only fragmented care.





Primary Care Platform: Personalized Primary Care Health Management

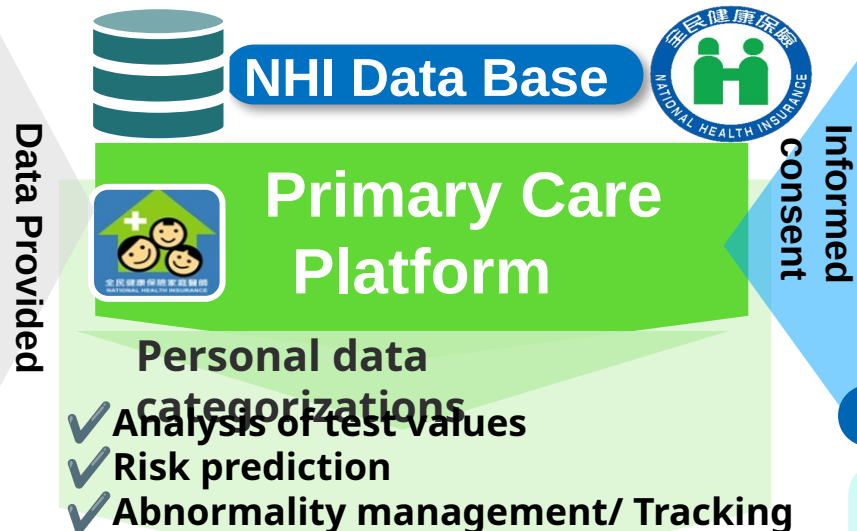
- ▶ Constructing **a personalized primary care health management platform** and data management framework
- ▶ Personal data aggregation and authorization management
- ▶ **Optimizing health data integration, risk prediction, and proactive alerts**
- ▶ Digitizing tracking management

Integrating personal health information



Data from the Ministry of Health and Welfare

Feedback to providers



Two-way communication, integrated counseling and education (via video)

Implementing holistic integrated care



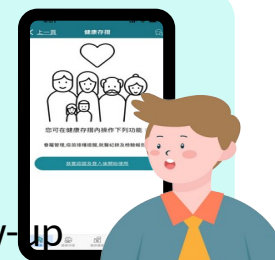
My Health Bank

- ✓ Feedback
- ✓ Screening reminder
- 1. Vaccine administration
- 2. Preventive healthcare
- 3. Abnormal screening results follow-up
- 4. Chronic prescription management

Public-private partnership
data value-added
applications

My Health Bank
SDK

Data integration
Innovative health services



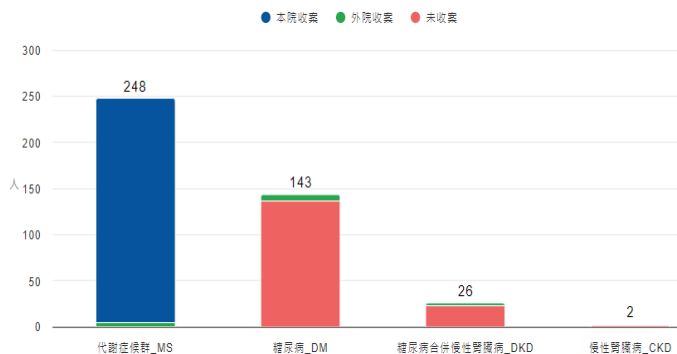
Family Doctor Platform: Basic Functional Architecture



Institutional management

- Visualizes overall case enrollment, including P4P-DM/CKD/DKD cases and follow-up progress
- Displays specific test results and comprehensive performance indicators

DM/CKD/DKD Enrollment Status

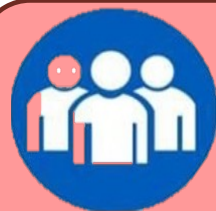
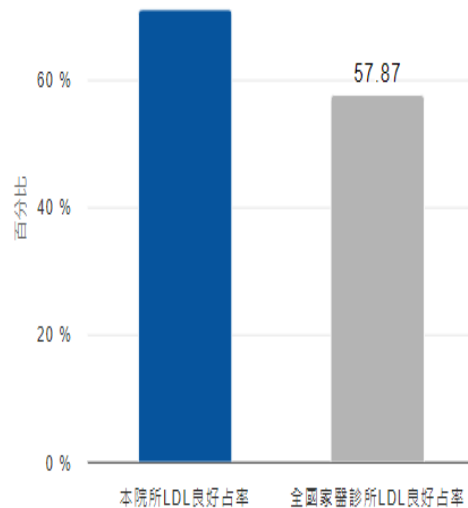


Performance of Indicators

評核指標名稱	執行狀況	得分
會員急診率	0.3281	1
會員固定就診率	0.5905	10
成人預防保健檢查	0.4083	7
子宮頸抹片檢查	0.2736	5

Three Highs Monitoring Data

Comparative analysis of LDL control rates between specific clinics and national averages



Case management

- Displays member health information, including:
 - ✓ Disease follow-up
 - ✓ Preventive health measures
 - ✓ Disease severity (ASCVD)

Preventive healthcare implementation status

項目	符合篩檢條件	篩檢結果	最後篩檢日	建議下次篩檢日期
成人預防保健	Y	-	2024/02/20	
子宮頸抹片檢查	N	Overview of health measures execution and patient follow-up		
糞便潛血檢查	Y		2023/04/15	2025/04/14
BC肝篩檢	Y		2023/04/15	

Disease severity classification

High risk

分級說明：就醫診斷糖尿病或慢性腎臟病或LDL-C檢驗數值大於190mg/dL。

關鍵數值：建議低密度膽固醇(LDL)控制在100mg/dL以下

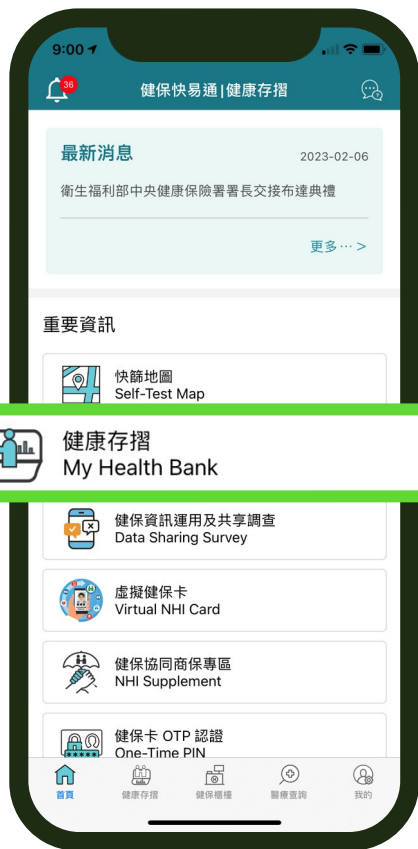
Test value trends over three years





MY HEALTH BANK: PERSONAL HEALTH RECORD(PHR)

The NHIA provides the insured with NHI data and related health checkup data through *My Health Bank*.



Filing Claims and NHI Card Data Uploading

Outpatient records from Western, Chinese and dental clinics, hospitalization records, medication records, surgery records, and allergy records



Encourage Medical Institutions to Upload

Tests and exams results, imaging/pathology tests and exams report, discharge summary, self-paid health checkup data



Data from Other Agencies

Consent to organ donation, consent to palliative care(MOHW); vaccination records (CDC); adult health checkup data, cancer screening data (HPA); *Kinmen Health Passbook* (Kinmen County Government)



Disease Risk Assessment

Liver cancer risk prediction, end-stage renal disease assessment and cardiovascular disease assessment

Disease Surveillance

Tracking patients with diabetes, early-stage chronic kidney disease, and hepatitis B&C

Self Reporting

Physiological measurement data, dietary/exercise behavioral indicators, menstrual cycle records, and self-paid vaccination records



Users: **11.49 million**

Number of uses: **394 million** (As of February 29, 2024)

AI-Assisted Digital Care

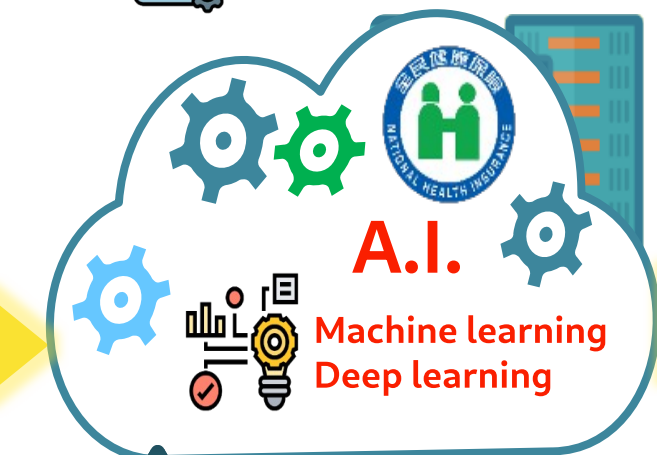
Data Entry into NHI database

- ✓ Gender
- ✓ Age
- ✓ Family Histories
- ✓ Nutrition
- ✓ Life Style
- ✓ Chronic Disease
- ✓ Lab Results
- ✓ Medications

- ✓ ICD-10 Code (aDCSI 7 categories of complications/ 52 subtypes)



NHI Cloud



Data Upload

Data Feedback



Medical facilities
Physicians



Patient

Data Output Prediction

Before illness
Risk of Diabetes

Onset

- 1) Within 10 years
- 2) Within 5 years
- 3) Within 1~3 years

After illness

Classification of Diabetes Complications(Sixth levels)

Key Indicators

- 1) Overall hospitalization rate
- 2) Hospitalization rate for specific causes
- 3) Mortality rate
- 4) Chronic disease prevalence
- 5) Medical expenses

Primary Care Platform Risk-Based Care and Payment

Risk levels

● : Low Risk

● : At Risk

● : Medium Risk

● : Medium-High Risk

● : High Risk

● : Severe Risk

Care diversion

Clinic
Public Health Center

District Hospital

Regional Hospital

Medical Center

Different incentives

Risk Prediction



Risk-Based Care and Payment



INTEGRATED, PATIENT-CENTERED, COMMUNITY-BASED HEALTH CARE

Patients with Mobility

“**Family Physician Program 2.0**,” will gradually integrate chronic disease care and preventive care.

- ✓ Initially intervened by the “P4P”
- ✓ Capitation including Chronic Disease Care



Metabolic
Syndrome Intervention

Prodromal Stage



P4P
for Chronic Disease
Care

Clinical Stage

Early Diagnosis, Early Treatment

Secondary Prevention

Patients with Disabilities

Integrated Home Health Care 2.0 :

Holistic Care: a community-based care model that not only provides home health care but also includes **home-based hospice care** and **acute care**, that also covers residents in long-term care facilities.

Care Plan For Residents in Long-Term Care Facilities :



Promoting Holistic Care

- ▶ Family physicians should have comprehensive health information for their patients, including disease care and preventive care.
- ▶ Providing patient-centered, integrated health care and long-term care.